

Dwight Alan Meglan, Ph.D.

Medical Device Technology Developer & Creator · **Surgical Robotics, Simulation & Image-Guided Systems**

FOUNDER · CHIEF TECHNOLOGY OFFICER · INVENTOR · TECHNICAL DILIGENCE & EXPERT WITNESS

Needham, MA

dmevlan@gmail.com

[linkedin.com/in/dmevlan](https://www.linkedin.com/in/dmevlan)

35+

YEARS BUILDING
MEDICAL DEVICES & SYSTEMS

15

COMMERCIAL SURGICAL
ROBOT PROGRAMS

90+

U.S. & INTERNATIONAL
PATENT FAMILIES

>\$15M

RESEARCH GRANTS
AUTHORED & WON

6+

VENTURES FOUNDED
MULTIPLE EXITS



CTO · surgical simulation



CTO · VIST simulator



HT Medical · medical VR & haptics



musculoskeletal simulation



Team 1757 lead mentor · 2008-present

**SIMULATION &
TRAINING**



Hugo · 3D-viz lead / fellow



robotic surgery



first NOTES robot



epicardial robot · founder



lung therapy · CTO



Ottava robotics



co-founder · architect



stroke robotics



CSP pacing guidance · founder & CTO



modular colonoscope · founder & CTO



consultancy · development · expert witness

MEDICAL ROBOTICS & DEVICES



iPlatform / Monarch



Spirus spiral endoscope · \$60M exit



insulin patch pump



endovascular robot · CTO



cardiac stimulation · CTO



Director, Medical Applications · MERL



orthopedic biomechanics fellow



Ph.D. · gait analysis laboratory



ASSERT Centre · visiting professor

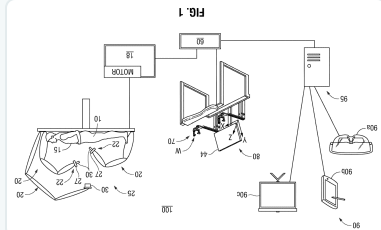
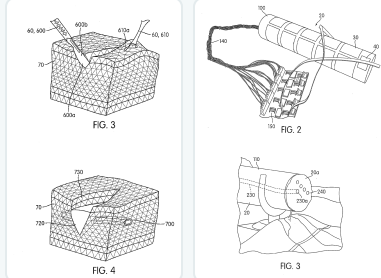
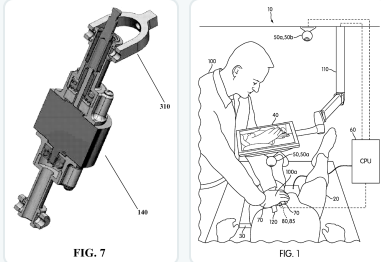
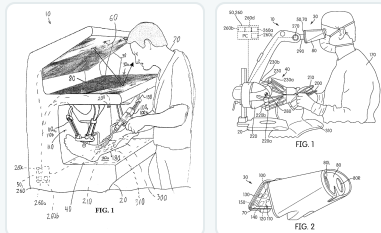


HeartLander origin license

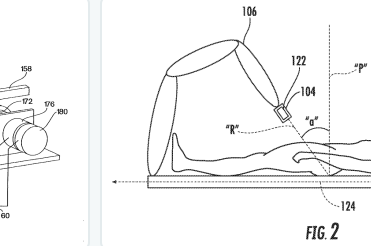
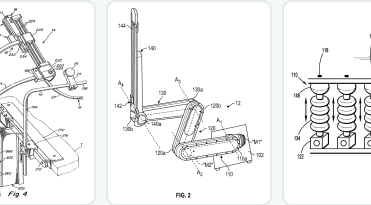
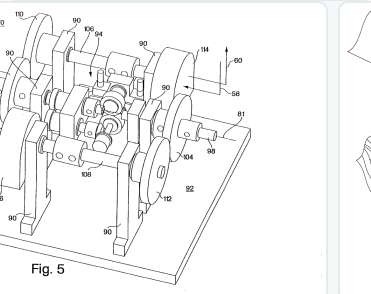
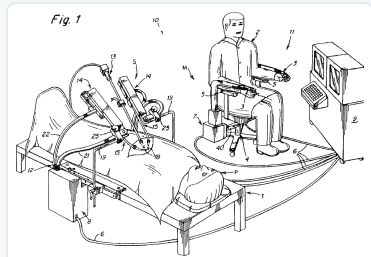
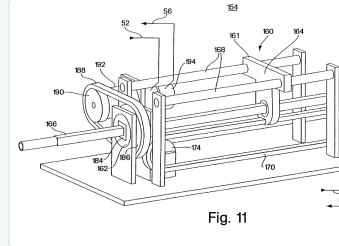
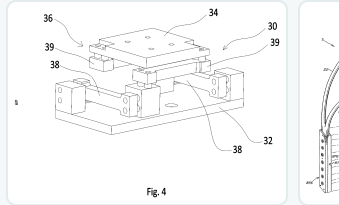
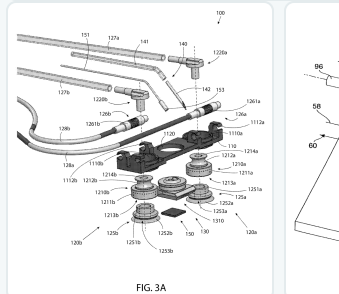
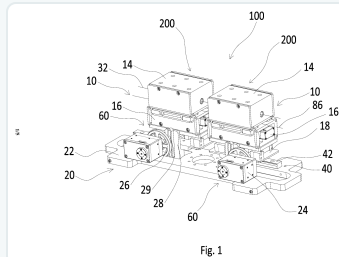


biomedical business development

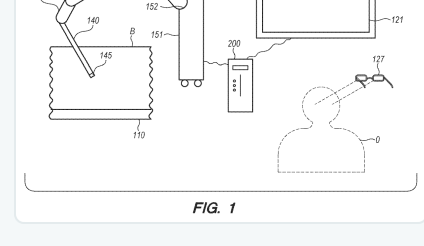
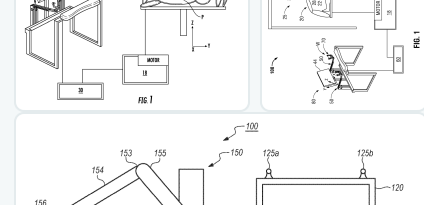
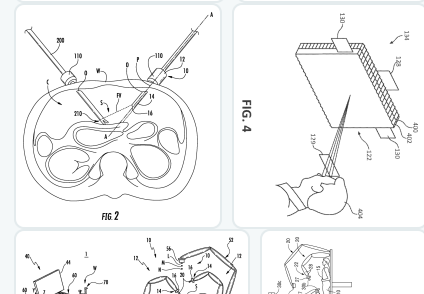
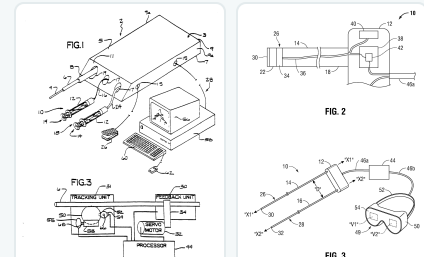
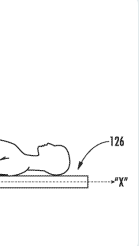
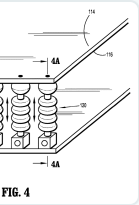
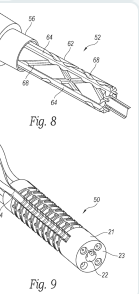
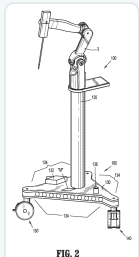
**RESEARCH & INDUSTRY
LABS**



SIMULATION & TRAINING



MEDICAL ROBOTICS & DEVICES



VISUALIZATION, SENSING & GUIDANCE

All drawings are from granted U.S. and international patents on which Dr. Meglan is an inventor: public records; each figure links to its patent on Google Patents.

03 Medical Device & Robotics Experience: Role & Contribution Matrix

DWIGHT A. MEGLAN, PH.D.

PERIOD	ORGANIZATION	ROLE	PROJECT / FOCUS	ARCHITECT	FOUNDER / CTO	DETAILED DESIGN	BUILD & TEST	PRODUCT / TRIAL	IP
2023-pres	Cardathea •	Founder & CTO	CSP pacing-lead placement guidance	✓	✓	✓	✓	✓	✓
2022-pres	Rocky Mountain Biphasic •	CTO	Novel cardiac / insulin stimulation	✓	✓	✓	✓	✓	✓
2022-pres	Elements Endoscopy •	Founder & CTO	Modular drive-by-wire colonoscope	✓	✓	✓	✓	✓	✓
2022-pres	Isola Therapeutics •	CTO	Aerosolized / fluidic lung therapy	✓	✓	✓	✓	✓	✓
2021-pres	Angio8 •	Founder & CTO	Endovascular catheterization robot	✓	✓	✓	✓		✓
2017-pres	Exemplar Devices •	Principal / Consultant	Device development • diligence • expert witness	✓		✓	✓		✓
2019-2023	Auris / J&J Robotics	Consultant	Ottava robot: full-arm contact sensorization	✓		✓	✓		✓
2013-2017	Medtronic (Covidien)	System Engineer Fellow	Hugo robot: 3D-visualization lead	✓	✓	✓	✓	✓	✓
2013-2019	Univ. College Cork	Visiting Research Prof.	Surgical-training research (ASSERT)						
2005-2025	HeartLander Surgical	Founder & CTO	Epicardial VT-therapy robot (NIH SBIR) • Cardathea predecessor	✓	✓	✓	✓	✓	✓
2010-2011	Spirus → Olympus	Independent Designer	Powered spiral-endoscope electronics	✓		✓	✓	✓	
2004-2013	SimQuest International	CTO	Surgical simulation (>\$14M in grants)	✓	✓	✓	✓	✓	✓
2003-2004	Foster-Miller	Biomed Bus. Dev. Lead	Medical-device practice expansion						
2002-2003	Hansen Medical	Chief System Architect	Telerobotic catheterization (co-founder)	✓	✓				✓
2001-2002	endoVia Medical	Sr. Program Manager	First endoluminal NOTES robot	✓	✓	✓	✓	✓	✓
1999-2001	Mentice	CTO	VIST interventional simulator	✓	✓	✓	✓	✓	✓
1996-1999	Mitsubishi (MERL)	Director, Medical Apps	Interventional-cardiology simulator	✓	✓	✓	✓	✓	✓
1994-1996	HT Medical → Immersion	Engineering Coordinator	Virtual surgical environments (VR)	✓	✓	✓	✓		✓
1993-1994	MusculoGraphics	Sr. Research Scientist	Musculoskeletal-simulation software	✓		✓	✓		
1991-1993	Mayo Clinic	Research Fellow	Orthopedic biomechanics	✓		✓	✓		
1989-1991	Ohio State University	Research Engineer	Gait-analysis laboratory			✓	✓		

• = active engagement. ✓ = direct, hands-on contribution in that capacity. Additional consulting / advisory: Neptune Medical (GI robotics) • Imperative Care (stroke robotics) • Abbott • Thermalin (insulin patch pump) • Dyme Medical (insulin delivery) • MIIS (imaging).

Inventor on **90+ U.S. and international patent and application families** spanning surgical robotics, simulation, endoscopy, imaging, and interventional cardiology; assignees include Medtronic, Intuitive / Hansen, Immersion, SimQuest, HeartLander, and Exemplar. Foundational grants include **US 6,096,004** (master/slave control of tubular medical tools, the basis of Hansen Medical) and **US 11,547,520** (robotic device & viewer-adaptive stereoscopic display). Patent numbers below link to the full records.

PATENT	TITLE	ASSIGNEE	PATENT	TITLE	ASSIGNEE
US 12,567,506	Simulation-based surgical procedure planning	Medtronic	US 11,529,038	Endoscope with IMU / haptics	Exemplar
US 12,491,028	Suturing guidance system	Medtronic	US 11,258,964	Camera-viewpoint synthesis in MIS	Medtronic
WO 2025/226661	Endoscope handle sterilizer	Exemplar	US 11,234,883	Operating table for robotic systems	Medtronic
US 12,440,286	Partial-reusable endo-tool robot	Exemplar	US 11,123,149	Angled-endoscope visualization	Medtronic
US 12,357,406	Robotic surgical collision detection	Medtronic	US 11,058,288	Amplifying surgical-region changes	Medtronic
US 12,272,463	Methods for surgical simulation	Medtronic	US 10,834,332	Camera viewpoints in MIS	Medtronic
US 12,262,863	Image mapping & fusion during surgery	Medtronic	US 10,660,714	Optical force sensor	Medtronic
US 12,185,908	Endoscope with IMU / haptic controls	Exemplar	US 9,607,528	Combined tissue & bone simulator	SimQuest
US 12,106,508	Mitigating collision of a robotic system	Medtronic	US 9,142,144	Hemorrhage-control simulator	SimQuest
US 12,102,288	Multi-use endoscopes	Exemplar	US 9,092,996	Microsurgery simulator	SimQuest
US 11,986,261	Surgical robotic-cart placement	Medtronic	US 8,915,743	Burr-hole drilling simulator	SimQuest
US 11,583,356	Controller for imaging device	Medtronic	US 8,671,950	Robotic medical instrument system	Hansen
US 11,553,984	Robotic system with embedded imager	Medtronic	US 7,959,557	Robotic medical instrument system	Hansen
US 11,548,140	Radio-based arm-cart location	Medtronic	US 7,699,835	Robotically controlled instruments	Hansen
US 11,547,520	Robotic device & stereoscopic display	Medtronic	US 6,096,004	Master/slave tubular medical tools	Mitsubishi

FURTHER GRANTED & PUBLISHED FAMILIES: A SELECTION

US 12,453,548 · US 12,310,687 · US 12,256,890 · US 12,193,765 · US 12,029,517 · US 12,004,773 · US 11,576,739 · US 11,517,183 · US 10,925,636 · US 10,779,856 · US 10,251,672 · EP 3,359,075 · WO 2024/051854 · WO 2021/242681 · WO 2021/133483 · WO 2021/067591 · WO 2019/204013 · WO 2019/204011 · WO 1998/010387 · US 2020/0184638 · US 2019/0088162 · US 2018/0310875 · US 2010/0178644 · US 2009/0011394 · US 2007/0123748

Plus 40+ additional U.S., European, Japanese, Chinese, and PCT filings across all domains; full list available on request, or via [Google Patents](#).

GRANTS & CONTRACTS AUTHORED: >\$15M AWARDED

Precise Minimally Invasive Treatment of Ventricular Tachycardia	NIH Fast-Track SBIR
Modular Flexible Semi-Disposable Endoscope Platform	NIH SBIR I
Drug Delivery by Endoesophageal Lavage for Esophageal Cancer	NCI SBIR I
HeartLander as an Epicardial Injection-Delivery System	NIH SBIR I
Open-Source Surgical Simulation for Combat Casualty Care	U.S. Army
Intracranial Hematoma / Burr-Hole & Trauma-Flap Simulator	Army SBIR I / II / III
AR System to Teach Hemostatic-Agent Use	NIH SBIR I
Simulation-Based Wound-Closure Training System	NIH SBIR Fast-Track
Haptics-Optional Surgical Training System	Army SBIR I & II
Simulation-Based Open-Surgery Training System	Army SBIR I & II
Simulation-Based Dental Training System	NIH SBIR I & II
Leg-Trauma Wound-Simulation Trainer for Army Medics	DARPA

RECOGNITION & SERVICE

Senior Member, IEEE (2009)

Senior Member, ACM (2009)

NIH Study-Section Reviewer

U.S. Defense SBIR Reviewer

Canadian NSERC Reviewer

ACS Steering Committee on Surgical Simulation (2001)

FIRST Robotics 1757 Lead Mentor: NE District Champions 2023 · Worlds top 4%

SIGNATURE ACHIEVEMENTS

- **\$60M Olympus exit:** sole developer of the UI & motor controls for Spirus Medical's powered spiral endoscope.
- **Medtronic Hugo:** 3D-visualization lead & technical fellow; low-latency stereoscopic imaging to the da Vinci standard.
- **First NOTES surgical robot:** built the first endoluminal master-slave telerobot; assets passed to Hansen & Intuitive.
- **Mentice VIST:** designer / lead engineer of the most successful surgical-simulation product; 400+ units, 20+ years on market.
- **Co-founder, Hansen Medical:** recruited by Intuitive founder Fred Moll; foundational catheter-robot IP (US 6,096,004).
- **Medtronic robotics ML group:** identified & advocated the acquisition (Digital Surgery).
- **OpenSurgSim:** conceived & won funding for the DoD open-source surgical-simulation platform.
- **Technical & IP due diligence:** advisor to investors and acquirers of surgical-robotics and simulation companies.
- **Expert witness:** medical-device & simulation litigation; patent & trade-secret assessment, reports, depositions.

SELECTED PUBLICATIONS

- Coles T, Meglan D, John N. "The Role of Haptics in Medical Training Simulators." *IEEE Trans. Haptics*, 2011;4(1):51-66.
- Meglan D, et al. "Development of an Open Surgery Simulator." *Medicine Meets VR*, 2011.
- Champion HR, Meglan DA. "Minimizing surgical error by incorporating objective assessment." *J. Am. Coll. Surg.*, 2008.
- Dawson SL, et al. "Computer-based simulator for interventional-cardiology training." *Catheter. Cardiovasc. Interv.*, 2000.